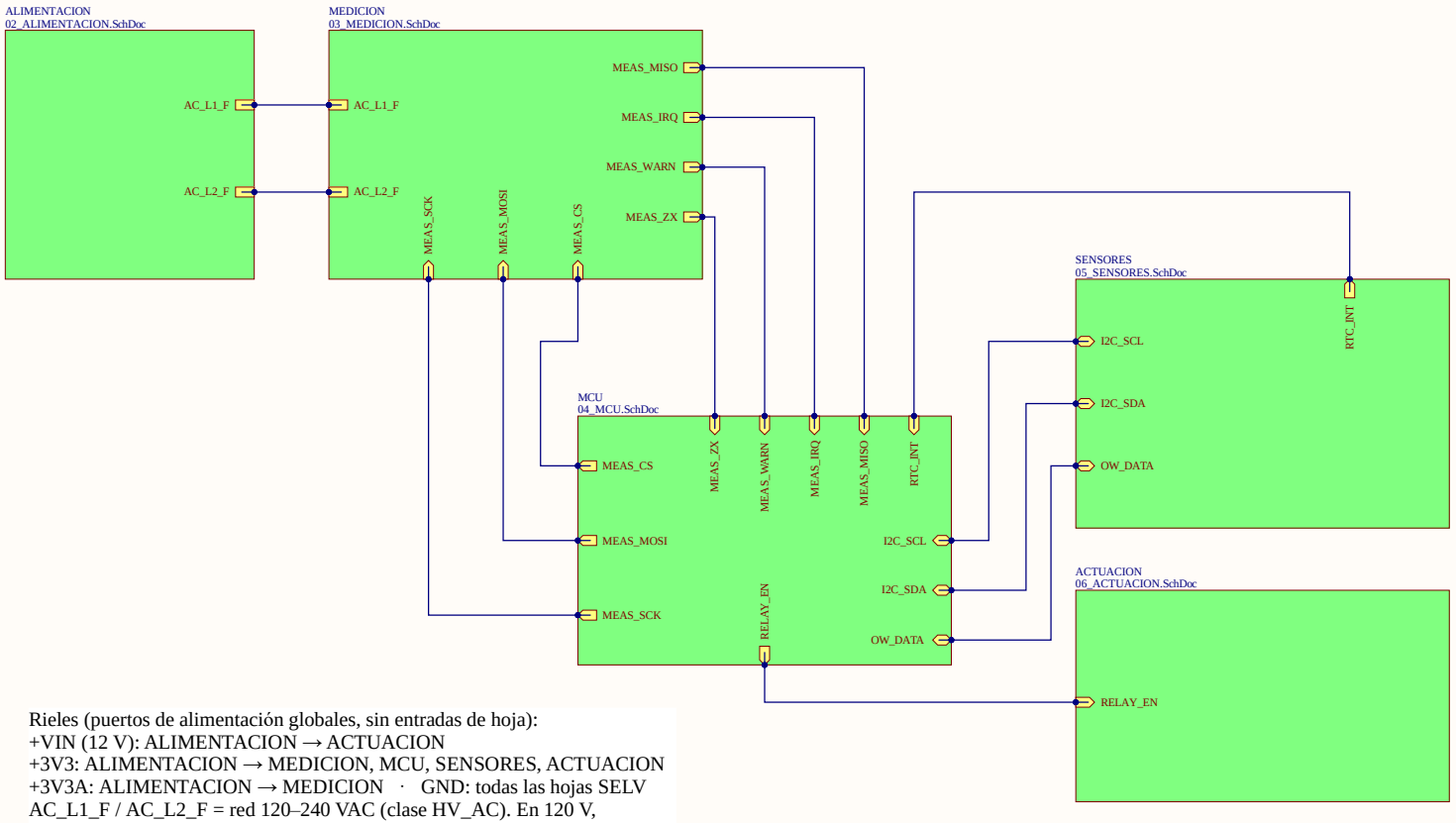
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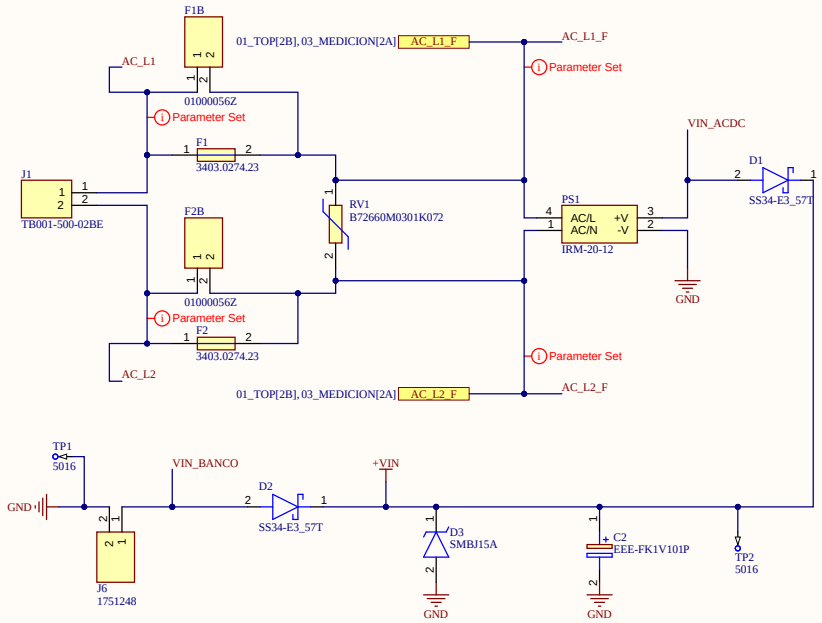
Rieles (puertos de alimentación globales, sin entradas de hoja):
+VIN (12 V): ALIMENTACION → ACTUACION
+3V3: ALIMENTACION → MEDICION, MCU, SENSORES, ACTUACION
+3V3A: ALIMENTACION → MEDICION · GND: todas las hojas SELV
AC_L1_F / AC_L2_F = red 120–240 VAC (clase HV_AC). En 120 V, AC_L2_F es el neutro.

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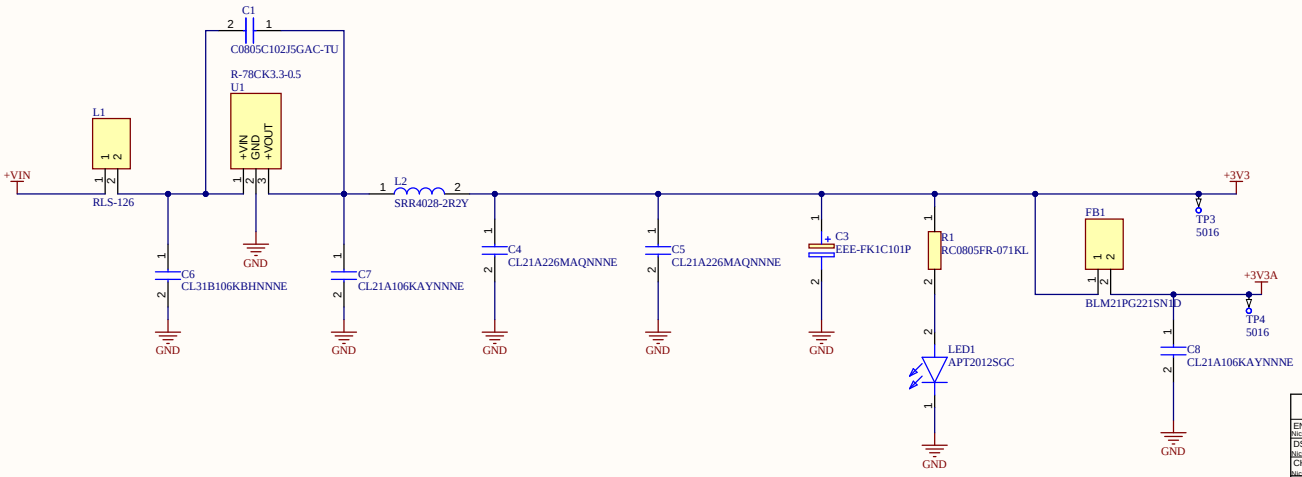
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Alimentar desde breaker exclusivo de 2-6 A (bipolar en 240 V). J1: 1 = L1/L, 2 = L2/N. J6: 12 V de banco (9–15 V).



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Parameter Set

01_TOP[2B], 02_ALIMENTACION[3A] AC_L1_F — R2 ERJ-P08F1004V — R3 ERJ-P08F1004V — R4 ERJ-P08F1004V — R5 ERJ-P08F1004V — HV_INP

HV_D1 HV_D2 HV_D3

R6 ERA-6AEB103V — C9 C0805C102J5GAC-TU

01_TOP[2B], 02_ALIMENTACION[3B] AC_L2_F — AC_L2_F

U2 AMC3330DWE

HV_DCDC 1 — DCDC_OUT — DCDC_IN — 16 AMC_LDO
AC_L2_F 2 — DCDC_HGND — DCDC_GND — 15
HV_DCDC 3 — HLDO_IN — DIAG — 14 AMC_DIAG
AC_L2_F 4 — NC — LDO_OUT — 13 AMC_LDO — +3V3
HV_LDO 5 — HLDO_OUT — VDD — 12
HV_INP 6 — INP — OUTP — 11 AMC_OUTP
AC_L2_F 7 — INN — OUTN — 10 AMC_OUTN
AC_L2_F 8 — HGND — GND — 9 — GND

HV_DCDC

C10 CL21B105KAFNNNE — C11 C0805C102J5GAC-TU — C12 CL21B104KBCNNNC — AC_L2_F

HV_LDO

C16 C3216NP01H104J160AA — C17 C0805C102J5GAC-TU — AC_L2_F

AMC_LDO — C13 CL21B104KBCNNNC — GND

+3V3 — C14 CL21B105KAFNNNE — C15 C0805C102J5GAC-TU — R7 RC0805FR-0747KL — AMC_DIAG — TP5 5016

Parameter Set

AMC_OUTP — C18 CL21A106KAYNNNE — R9 ERA-6AEB203V — ATM_VP

R10 ERA-6AEB133V — C19 C0805C472J5GACTU

AMC_OUTN — C21 CL21A106KAYNNNE — R12 ERA-6AEB203V — ATM_VN

R14 ERA-6AEB133V — C22 C0805C472J5GACTU

CT_P — R8 RC0805FR-071KL

J3 SJ1-3515-SMT-TR — TIP — TIP_SWITCH — RING — RING_SWITCH — SLEEVE

R11 ERA-6AHD100V — ATM_IP — C20 C1206C333J5GACTU

D4 SMAJ5.0CA

R13 ERA-6AHD100V

R15 RC0805FR-071KL — ATM_IN — C23 C1206C333J5GACTU

CT_N

ATM_CF2 — TP6 5016

U3 ATM90E26-YU-R

1	MMD1	MMD0	28 +3V3
2	DGND	SDI/URX	27 MEAS_MOSI — 01_TOP[3B], 04_MCU[1B]
3	DVDD	SDO/UTX	26 MEAS_MISO — 01_TOP[3A], 04_MCU[1B]
4	RESET	SCLK	25 MEAS_SCK — 01_TOP[3B], 04_MCU[1B]
5	AVDD	CS	24 MEAS_CS — 01_TOP[3B], 0
6	AGND_1	OSCO	23
7	I2P	OSCI	22 XTAL_O
8	I2N	ZX	21 MEAS_ZX — 01_TOP[3B], 04_MCU[1B]
9	RESV_LOW	IRQ	20 MEAS_IRQ — 01_TOP[3B], 04_MCU[1B]
10	I1P	CF2	19 XTAL_I
11	I1N	CF1	18 ATM_CF1
12	USEL	WARNOUT	17 MEAS_WARN — 01_TOP[3B], 04_MCU[1B]
13	VREF	VP	16 ATM_CF2
14	AGND_2	VN	15 ATM_VP

GND +3V3 — ATM_RST — +3V3A — ATM_IP — ATM_IN — ATM_VREF

C24 CL21A106KAYNNNE — C25 CL21B104KBCNNNC

C28 CL21B104KBCNNNC

R16 RC0805FR-0710KL — ATM_RST — C29 CL21B104KBCNNNC

XTAL_I — Y1 ECS-81.92-12-30Q-EN-TR — XTAL_O

C26 C0805C120J5GACTU — C27 C0805C120J5GACTU

C30 CL21B105KAFNNNE — C31 C0805C102J5GAC-TU

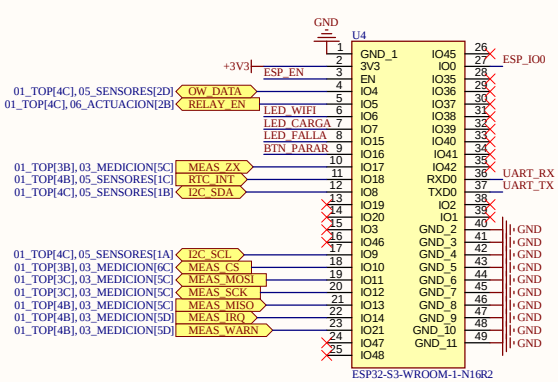
ATM_CF1 — R17 RC0805FR-071KL — LED2 APT2012SGC

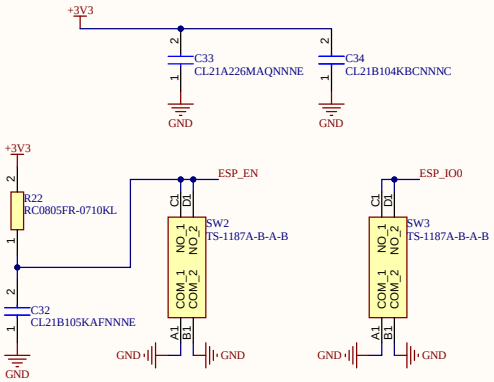
R18 RC0805FR-0710KL — MEAS_CS

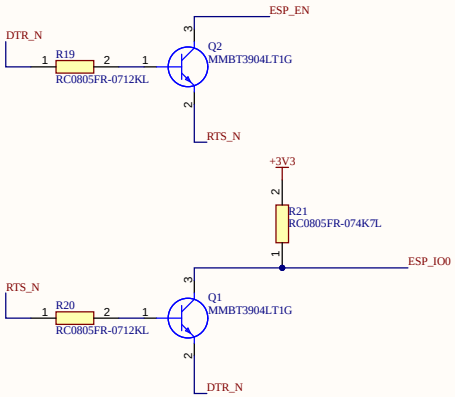
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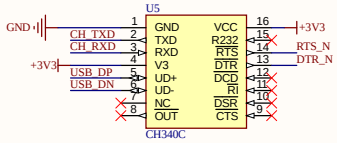
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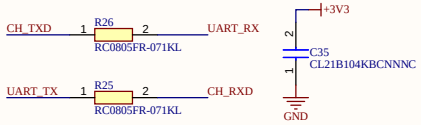
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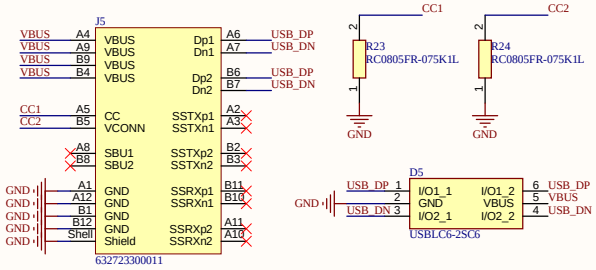


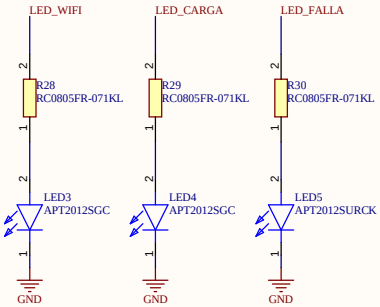


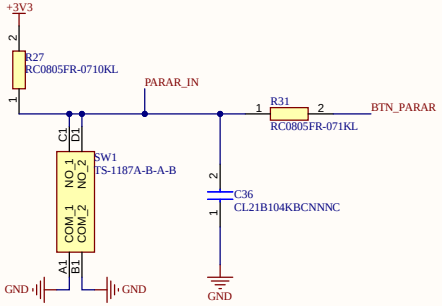


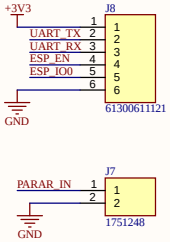








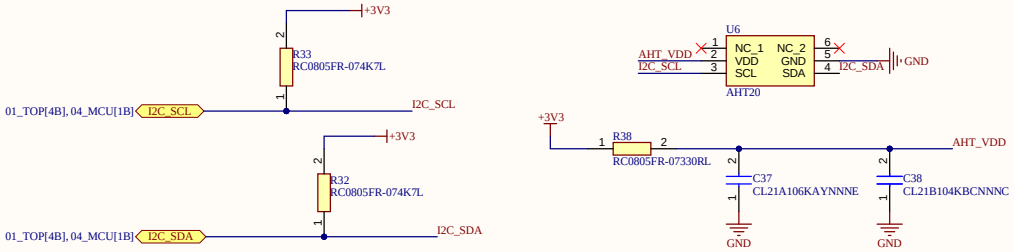


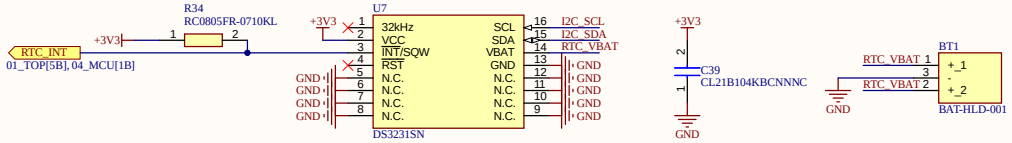


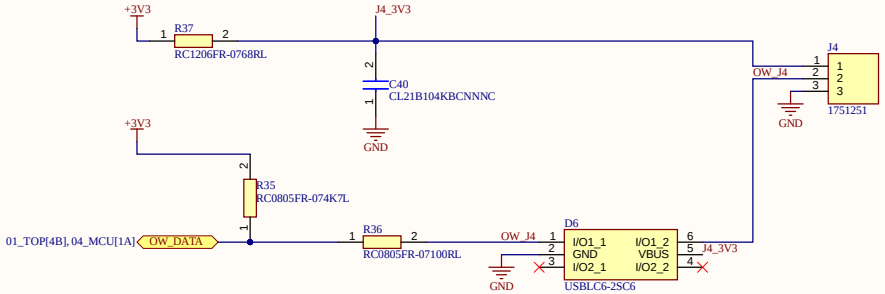
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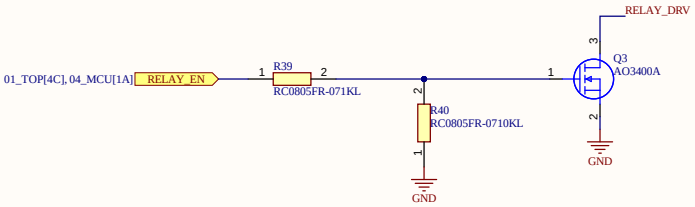


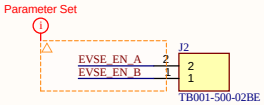


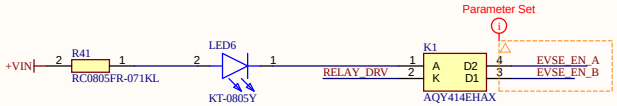
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PROJECT REVISION: 0.1

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Comment	Description	Designator	Footprint	LibRef	Quantity
BAT-HLD-001	Battery holder CR2032, SMD	BT1	BATHLD001	BAT-HLD-001	1
C0805C102J5GAC-TU	Capacitor	C1, C9, C11, C15, C17, C31	CAPC2012X88N	C0805C102J5GAC-TU	6
EEE-FK1V101P	Capacitor Polarised	C2	EEHZT1E331UP	EEE-FK1V101P	1
EEE-FK1C101P	Capacitor Polarised	C3	EEE1AA101SP	EEE-FK1C101P	1
CL21A226MAQNNNE	Capacitor	C4, C5, C33	CAPC2012X140N	CL21A226MAQNNNE	3
CL31B106KBHNNNE	Capacitor	C6	CAPC3216X180N	CL31B106KBHNNNE	1
CL21A106KAYNNNE	Capacitor	C7, C8, C18, C21, C24, C37	CAPC2012X145N	CL21A106KAYNNNE	6
CL21B105KAFNNNE	Capacitor	C10, C14, C30, C32	CAPC2012X88N	CL21B105KAFNNNE	4
CL21B104KBCNNNC	Capacitor	C12, C13, C25, C28, C29, C34, C35, C36, C38, C39, C40	CAPC2012X88N	CL21B104KBCNNNC	11
C3216NP01H104J160AA	Capacitor	C16	CAPC3216X180N	C3216NP01H104J160AA	1
C0805C472J5GACTU	Capacitor	C19, C22	CAPC2012X88N	C0805C472J5GACTU	2
C1206C333J5GACTU	Capacitor	C20, C23	CAPC3216X180N	C1206C333J5GACTU	2
C0805C120J5GACTU	Capacitor	C26, C27	CAPC2012X88N	C0805C120J5GACTU	2
SS34-E3_57T	Schottky Diode	D1, D2	DIOM7959X262N	SS34-E3_57T	2
SMBJ15A	TVS Diode (Uni-directional)	D3	DIOM5436X244N	SMBJ15A	1
SMAJ5.0CA	TVS Diode (Bi-directional)	D4	DIONM5026X229N	SMAJ5.0CA	1
USBLC6-2SC6	Integrated Circuit	D5, D6	SOT95P280X145-6N	USBLC6-2SC6	2
3403.0274.23	Fuse	F1, F2	3403027423	3403.0274.23	2
01000056Z	Undefined or Miscellaneous	F1B, F2B	01000056Z	01000056Z	2
BLM21PG221SN1D	Ferrite Bead	FB1	BEADC2012X105N	BLM21PG221SN1D	1
TB001-500-02BE	Connector	J1, J2	TB001-500-02BE	TB001-500-02BE	2
SJI-3515-SMT-TR	Connector	J3	SJI3515SMTTR	SJI-3515-SMT-TR	1
1751251	Connector	J4	1751251	1751251	1
632723300011	Conn USB 3.1 Type C RCP 24 POS 0.5mm/0.4mm Solder RA SMD/Thru-Hole 24 Terminal 1 Port T/R	J5	632723300011	CMP-1502-03214-4	1
1751248	Connector	J6, J7	1751248	1751248	2
61300611121	Connector	J8	HDRV6W79P0X254_1X6_1524X254X894P	61300611121	1
AQY414EHAX	PhotoMOS 1 Form B (NC), 400 V 120 mA, reinforced insulation 5000 Vrms, SMD DIP-4	K1	SOP254P962X315-4N	AQY-414EHAX	1
RLS-126	Power Supply	L1	RLS126	RLS-126	1
SRN4028-2R2Y	Inductor	L2	SRN4028	SRN4028-2R2Y	1
APT2012SGC	LED	LED1, LED2, LED3, LED4	LEDC2012X85N	APT2012SGC	4
APT2012SURCK	LED	LED5	LEDC2012X85N	APT2012SURCK	1
KT-0805Y	LED yellow 0805, 590 nm	LED6	LEDC2012X85N	KT-0805Y	1
IRM-20-12	Power Supply	PS1	IRM2012	IRM-20-12	1
MMBT3904LT1G	Transistor BJT NPN	Q1, Q2	SOT96P237X111-3N	MMBT3904LT1G	2
AO3400A	MOSFET N-channel 30V 5.7A, Rds(on) 48mR max at Vgs 2.5V, SOT-23	Q3	SOT96P237X111-3N	AO3400A	1
RC0805FR-071KL	Resistor	R1, R8, R15, R17, R25, R26, R28, R29, R30, R31, R39, R41	RESC2012X60N	RC0805FR-071KL	12
ERJ-P08F1004V	Resistor	R2, R3, R4, R5	ERJP08_(1206)	ERJ-P08F1004V	4
ERA-6AEB103V	Resistor	R6	RESC2012X60N	ERA-6AEB103V	1
RC0805FR-0747KL	Resistor	R7	RESC2012X60N	RC0805FR-0747KL	1
ERA-6AEB203V	Resistor	R9, R12	RESC2012X60N	ERA-6AEB203V	2
ERA-6AEB133V	Resistor	R10, R14	RESC2012X60N	ERA-6AEB133V	2
ERA-6AHD100V	Resistor	R11, R13	RESC2012X60N	ERA-6AHD100V	2
RC0805FR-0710KL	Resistor	R16, R18, R22, R27, R34, R40	RESC2012X60N	RC0805FR-0710KL	6
RC0805FR-0712KL	Resistor	R19, R20	RESC2012X60N	RC0805FR-0712KL	2
RC0805FR-074K7L	Resistor	R21, R32, R33, R35	RESC2012X60N	RC0805FR-074K7L	4
RC0805FR-075K1L	Resistor	R23, R24	RESC2012X60N	RC0805FR-075K1L	2
RC0805FR-07100RL	Resistor	R36	RESC2012X60N	RC0805FR-07100RL	1
RC1206FR-0768RL	Resistor	R37	ERJP08_(1206)	RC1206FR-0768RL	1
RC0805FR-07330RL	Resistor	R38	RESC2012X60N	RC0805FR-07330RL	1
B72660M0301K072	Varistor	RV1	CU4032	B72660M0301K072	1
TS-1187A-B-A-B	Tact switch 5.1 x 5.1 x 1.5 mm, 160 gf, 50 mA 12 VDC	SW1, SW2, SW3	TS1187ABAB	TS-1187A-B-A-B	3
5016	Test Point, 1-Pin SMT, RoHS, Tape and Reel	TP1, TP2, TP3, TP4, TP5, TP6	KEYSTONE-5016_V	CMP-2000-05178-1	6
R-78CK3.3-0.5	Regulador conmutado 3.3 V 0.5 A, SIP3	U1	R78CK3305	R-78CK3.3-0.5	1
AMC3330DWE	Integrated Circuit	U2	SOIC127P1030X265-16N	AMC3330DWE	1
ATM90E26-YU-R	Integrated Circuit	U3	SOP65P778X199-28N	ATM90E26-YU-R	1
ESP32-S3-WROOM-1-N16R2	Integrated Circuit	U4	ESP32S3WROOM1N16R2	ESP32-S3-WROOM-1-N16R2	1
CH340C	USB to UART bridge	U5	SOIC127P600X175-16N	CH340C	1
AHT20	Integrated Circuit	U6	AHT20	AHT20	1
DS3231SN	I2C RTC with integrated TCXO and crystal, VBAT, SO-16W	U7	SOIC127P1030X265-16N	DS3231SN	1
ECS-81.92-12-30Q-EN-TR	Crystal or Oscillator	Y1	ECSB1921230QENTR	ECS-81.92-12-30Q-EN-TR	1